

1 The axioms we actually have: PA and ZF

Which axioms does the posterior pick up for arithmetic and set theory? Sources: track pa; track experiments for E2, E3(b), E6 and Props X9–X11; AS (Claude (Anthropic), 2026) is cited by its own numbering. Units are bits; proofs and full tables are in Appendix A. Three caveats hold throughout. (i) Track pa computes with the two-part code $L1^{\text{sch}}$ or with L0 under a learned grammar, never with the generative L1 of ??; its numbers are not L1 results. (ii) Candidate sets are hand-picked and finite. (iii) Derivation costs are those of explicit derivations, hence upper bounds. “Proved; computed” means a proof plus a derivation checked line by line by track pa’s checker `nd.py` (Appendix A.1).

1.1 Setting

Definition 1.1 (The two-part derivation code $L1^{\text{sch}}$). pa Defs 0.2, 0.3 Derivations are in a sound and complete natural-deduction calculus ND (??). An instance $f(\varphi)$ of a schema $f(P)$ is derived schematically in P and costs $\text{bits}(\pi) + L_Q(\varphi)$, the motive being coded once by the grammar. Per line, $\text{bits}(\pi)$ charges $\log_2 10$ for the rule, $\log_2 |T|$ per axiom citation, $\log_2(\text{line index})$ per premise reference and β per *written symbol*. The *decodable* variant also codes the number of lines, the premise counts of tc lines and the variables of subst and $\exists I$ lines. $L1^{\text{naive}}$ writes every formula out at the instance. On a finite candidate set the prior is $\pi(T) \propto 2^{-\beta \sum_{\tau \in T} |\tau|}$: $\beta = \log_2 23 \approx 4.52$ bits per symbol of `nd.size` (a quantifier and its variable count 2) in the derivation computations, $\beta = 5$ per symbol of the `dtlib` size (a de Bruijn $\forall$ counts 1) in those built on AS’s code `u7`. $CL(T; D)$ is the total code length, ℓ_{cite} the cost of a one-line citation.

The decodable variant adds 4.8–7.3% to derivation costs and is a prefix code, so Kraft holds (proved): $L1^{\text{sch}}$ is the two-part form of a sub-probability likelihood, of the $L1^\sigma$ kind, not of plain L1 (??). The grammar Q is *learned*: each metavariable body is coded at its read-off position by sequential KT in the context (sort, depth, parent, child index), over 9 formula symbols ($=, \neg, \wedge, \vee, \rightarrow, \forall, \exists, \top, \perp$) and 14 term symbols. Sequential KT is a Dirichlet($\frac{1}{2}$) marginal (Krichevsky and Trofimov, 1981), as are the weights, so CL differences in the L0 computations are exact $\log_2$ posterior odds, except where templates overlap (Remark A.7).

Definition 1.2 (Refutation rule R_d).

pa Def 0.4 With data D and negative data N , R_d sets the likelihood of T to 0 if $T \vdash_{\leq d} \neg s'$ for some $s' \in D$ or $T \vdash_{\leq d} s$ for some $s \in N$ (Th_d in derivation size; under L0, $d = 0$ and “derives” means “has as an instance”). R_∞ is not computable.

1.2 Equivalent axiomatisations: usage decides

Let $\text{Dlt} := \forall u \forall v (u < v \leftrightarrow \exists z (u + Sz = v))$. For a motive φ in x (parameters allowed), $\text{Ind}(\varphi) := \varphi(0) \wedge \forall x (\varphi \rightarrow \varphi(Sx)) \rightarrow \forall x \varphi$; $\text{CVI}(\varphi) := \forall x (\theta_\varphi \rightarrow \varphi) \rightarrow \forall x \varphi$ with $\theta_\varphi := \forall y (y < x \rightarrow \varphi(y))$; $\text{LNP}(\varphi) := \exists x \varphi \rightarrow \exists x (\varphi \wedge \forall y (y < x \rightarrow \neg \varphi(y)))$. For nonempty $S \subseteq \{\text{Ind}, \text{CVI}, \text{LNP}\}$ let $\text{PA}_S := Q + \text{Dlt} + \{f(P) : f \in S\}$, $\text{PA}_f := \text{PA}_{\{f\}}$, and PA_{all} with all three.

Proposition 1.3 (Three forms of induction). *[proved; computed]*

pa Prop 1.1 With $B := \{Q1, \dots, Q5, \text{Dlt}\}$, for every motive φ : (A) $B \vdash \text{CVI}(\varphi) \rightarrow \text{Ind}(\varphi)$; (B) $B \vdash \text{Ind}(\theta_\varphi) \rightarrow \text{CVI}(\varphi)$; (C1) $\vdash \text{CVI}(\neg \varphi) \rightarrow \text{LNP}(\varphi)$; (C2) $\vdash \text{LNP}(\neg \varphi) \rightarrow \text{CVI}(\varphi)$. So PA_{Ind} , PA_{CVI} , PA_{LNP} have the same theorems.

theory	forms derived	lines	symbols	overhead	decodable
PA _{Ind}	CVI, LNP	63, 82	204, 332	1468.0, 2234.1	1551.2, 2341.8
PA _{CVI}	Ind, LNP	39, 20	127, 117	880.4, 646.4	945.1, 676.9
PA _{LNP}	Ind, CVI	65, 27	253, 128	1692.1, 757.5	1792.3, 799.1

Table 1: Overhead in bits per use of deriving a non-primitive form $f(P)$ over citing it ($L1^{\text{sch}}$, $\beta = \log_2 23$); a citation costs 15.5–15.8 bits; template priors Ind 72, CVI 81, LNP 86 bits. Computed; upper bounds (pa §1.2).

The derivations were checked schematically in P , replayed on 200 random motives, and re-checked by the referee’s independent checker (Appendix A.2).

Remark 1.4 (Patterns). [proved]

pa Rem 1.2 $CVI(P), LNP(P) \in \text{PAT}$, while $T_{\text{Ind}} \in \text{DT}^\circ \setminus \text{PAT}$ ($P(0)$, $P(Sx)$). So PA is Q, Dlt and one higher-order pattern (compare AS Prop 5.18: Q plus the first-order pattern IndEq).

Usage. Let the data be i.i.d. uses: a form f with probability p_f and a motive $\varphi \sim Q$, the datum being $f(\varphi)$. In PA_S a use costs $c_S(f) + L_Q(\varphi)$, $c_S(f)$ the cost of the cheapest derivation of $f(P)$ in a fixed library of checked derivations (a citation if $f \in S$; Table 1).

Proposition 1.5 (Usage limit). [proved, given the library constants]

pa Prop 1.3 $\frac{1}{n}(\text{CL}(PA_S; D_n) - \text{CL}(PA_{S'}; D_n)) \rightarrow \sum_f p_f(c_S(f) - c_{S'}(f))$ a.s.; the posterior concentrates on the S minimising $\mathbb{E}_p[c_S(f)]$ if it is unique.

Proposition 1.6 (Lower bound). [proved]^{pa Prop 1.4} Every $L1^{\text{sch}}$ -derivation of $CVI(P)$ from PA_{Ind} costs at least $\log_2 10 + 11\beta \approx 53$ bits more than citing it in PA_{CVI} ; every derivation of $\text{Ind}(P)$ from PA_{CVI} at least $\log_2 10 + 9\beta \approx 44$ bits more than citing it in PA_{Ind} .

The first version’s argument (“at least two lines”) was wrong, since a citation also writes $P(x)$; the proof in Appendix A.2 follows the premises back to a line that writes the form’s skeleton or antecedent.

Remark 1.7 (The usage-closed axiomatisation). [computed]^{pa §1.2} For $L1^{\text{sch}}$ with this library: (i) PA_{Ind} beats PA_{CVI} iff $p_{\text{CVI}}/p_{\text{Ind}} < 0.600$, and PA_{LNP} iff $p_{\text{LNP}}/p_{\text{Ind}} < 0.757$ (0.609, 0.765 under the decodable code; under $L1^{\text{naive}}$ the first ratio is 0.671–1.052, so the ratios are code-specific). (ii) PA_{all} pays 81 + 86 prior bits and 0.29 bits per citation more than PA_{Ind} , so it is ahead after about one use of a non-primitive form (in expectation after 1–12 schema uses for the mixes tried). (iii) So the posterior concentrates on the *usage-closed* axiomatisation: every form used at a rate above about $5 \cdot 10^{-4}$ per use is primitive. All candidates are PA-equivalent, so the theorems are right whichever wins (compare ??).

Proposition 1.8 (Recursion conventions). [proved; computed]

pa Prop 1.5 $Q + T_{\text{Ind}}$ and the theory with $Q4$, $Q5$ replaced by $0 + x = x$, $Sx + y = S(x + y)$ prove each other’s recursion laws with one induction each, so they are equivalent; the per-use overheads are 213.8 and 593.4 bits one way, 213.8 and 545.1 the other. The posterior adopts the convention the data use.

Remark 1.9 (ZF). [proved; computed; known; conjecture (reverse overheads)]^{pa §1.4} Separation from Replacement, Replacement from Collection and Foundation from $\in$ -induction are checked, at 574–1432 bits per use against template priors of 72–185 bits (Table 5). Collection from Replacement

holds over ZF, fails without Power Set (Zarach, 1996; Gitman et al., 2016) (known) and is open to us without Foundation; EInd from Found needs transitive closures (known; Kaye and Wong, 2007). The reverse overheads were not formalised (conjecture: much larger). So the textbook ZF axioms win exactly when they are used directly.

Remark 1.10 (E6: commutativity). [computed]^{experiments E6} Among eight forms of commutativity (??), the chain L1 identifies the generating form by $n = 32$ in every run, both among logically equivalent forms and among instance-equivalent but weaker ones; under L1_{cit}^{sel} five closed-guard forms tie, and the mass on provers of $\forall x \forall y x + y = y + x$ tends to their prior share 0.088 (??).

1.3 The MDL finding in PA

AS found that “MDL tracks the statistics of usage, not the logical boundaries of schemas” (AS §6.10, Props E.16, E.17, Table 12); its codes PC and DPC are sequential KT, so its table already gives $\log_2$ posterior odds. Track pa re-ran AS’s code u7 (reproducing AS Table 12 at $n = 1000$) on $T^* := Q + T_{\text{Ind}}$ against the root split H_{root} (7 templates T_f) and the split H_{used} into the roots F that occur, adding the grammars SDPC (one learned positional grammar shared by all templates), RDPC (T^* ’s grammar may condition on the motive’s root) and CF (one nonterminal per sort), under usage laws G1 (u7’s natural law, 7 roots), G2 and G3 (4 and 3 roots).

Proposition 1.11 (Chain rule). [proved]^{pa Prop 2.1} *With fixed parameters, if T^* has weight w on T_{Ind} and body law $\mathcal{Q}(\varphi) = \mathcal{Q}(f)\mathcal{Q}(\varphi \mid f)$, f the root, then H_F ($F \supseteq$ the roots in the support) with weights $w\mathcal{Q}(f)$ on T_f and body laws $\mathcal{Q}(\cdot \mid f)$ has $P_{H_F} = P_{T^*}$.*

Proposition 1.12 (Occam identity). [(a), (c) proved; (b) proved under its growth condition; computed]^{pa Prop 2.3} *Use L0 with a Dirichlet($\frac{1}{2}$) citation index and a shared learned positional grammar (KT per context c , alphabet A_c); $\text{KT}(k; A)$ is $-\log_2$ of the Dirichlet($\frac{1}{2}$) marginal of counts k over A symbols. Suppose every datum is covered by exactly one template of H and one of H' , and, for a set C of contexts, H' codes datum by datum exactly the (context, symbol) pairs that H codes outside C . (a) $\text{CL}(H') - \text{CL}(H) = \Delta_{\text{prior}} + \text{KT}(n'; |H'|) - \text{KT}(n; |H|) - \sum_{c \in C} \text{KT}(m_c; |A_c|)$ (n, n' citation counts, m_c symbol counts in c). (b) If all nonzero counts grow linearly, $\text{KT}(k; A) = N\hat{H}(k) + \frac{A-1}{2} \log_2 N + O(1)$, $N = \sum k$. (c) If $C = \{c\}$, the data visiting c are those of one template of H , and H' ’s template and the pair (H ’s template, symbol in c) determine each other, then $\text{CL}(H') - \text{CL}(H) = \Delta_{\text{prior}} + \frac{(|H'| - |H|) - (|A_c| - 1)}{2} \log_2 n + O(1)$; for a split of T_{Ind} into roots F under SDPC, $\Delta_{\text{prior}} + \frac{|F| - 9}{2} \log_2 n + O(1)$.*

The identity equals the computed SDPC code length to 0.1 bit at every n and law (slopes $-1, -2.5, -3$ bits per doubling for $|F| = 7, 4, 3$). It reconciles the signs of the Occam drift: with a fixed $\mathcal{Q}$ only the split pays Occam terms (??; E2: about 4 bits per doubling); with a learned grammar the schema also pays $\frac{1}{2} \log_2 n$ per formula symbol never used at the root.

Remark 1.13 (Refuted: a constant margin). [refuted]

^{pa §2.2; referee m2} The first version called the SDPC margin of H_{root} constant ($+770 \rightarrow +762$ bits for $n = 10^3 \rightarrow 2.56 \cdot 10^5$). It falls by 1 bit per doubling (Proposition 1.12(c), $|F| = 7$); extrapolated crossover $n \approx 2^{780}$.

On G1 at $n = 256000$ the split leads by 274656, 94089, 163028 and 17242 bits under NAIVE, PC, DPC and CF, and trails by 762 (SDPC) and 5986 (RDPC) bits (Table 6, Appendix A.3). So the finding is about the grammar, not the likelihood family: a split pays off linearly whenever $\mathcal{Q}$ ignores a feature that the split conditions on and that usage depends on; SDPC and RDPC hold it at the prior margin up to Occam terms. A derivation likelihood changes the consequences:

theorem s	$ s $	lines	$D_T(s)$	$\beta s + \ell_{\text{cite}}$	ratio	$\beta^*(s)$
$\forall x \neg Sx = x$	7	11	224.6	38.2	5.9	31.2
$\forall x 0 \cdot x = 0$	7	14	281.3	38.2	7.4	39.3
$\forall x 0 + x = x$	7	11	220.1	38.2	5.8	30.5
$\forall a \forall v Sa + v = S(a + v)$	13	23	576.0	65.3	8.8	43.8
$\forall q \forall x x + q = q + x$	11	55	1358.6	56.3	24.2	122.9
$\forall a \forall b \forall x (a + b) + x = a + (b + x)$	17	27	793.4	83.4	9.5	46.3

Table 2: Deriving versus memorising in $\mathbf{Q} + T_{\text{Ind}}(\mathbf{L1}^{\text{sch}}, \beta = \log_2 23 \approx 4.52; \text{bits})$: memorising wins at the first occurrence for every theorem. Computed; checked derivations; upper bounds (pa §3.2; more in Appendix A.4).

Proposition 1.14 (Detours). *[(a) proved; computed; (b) known]*

pa Prop 2.2 (a) A split H_F derives every induction instance if F contains a non-atomic connective, through the wrappers $\neg\neg P, P \wedge P, P \vee P, (0 = 0) \rightarrow P, \forall zP, \exists zP$ (six checked derivations; under $\mathbf{L1}^{\text{sch}}$ they cost 369.2–488.9 bits more than a citation). (b) If $F = \{=\}$, $\mathbf{Q} + H_F \subseteq \text{IOpen}$, which does not prove the irrationality of $\sqrt{2}$ (Shepherdson, 1964) (secondary sources only), while PA does.

So under $\mathbf{L1}$ MDL’s incompleteness for schemas (AS App. E.7) disappears once a split has a non-atomic root; the usage effect and the unsoundness error (Remark 1.32) remain (qualification (i) of ??).

1.4 Theorem data: statements given without proof

Hänni proposed that “statements given without proof in our data set should be fairly easily derived from axioms”. Then the data are theorems. A theory T can derive a datum s , at cost $D_T(s)$ (cheapest library derivation under $\mathbf{L1}^{\text{sch}}$), or adopt it as a ground axiom.

Proposition 1.15 (Memorisation threshold). *[proved]*

pa Prop 3.1 If s occurs r times and the other data have the same best derivations under T and $T \cup \{s\}$, then $\text{CL}(T \cup \{s\}) - \text{CL}(T) = \beta|s| + \Delta I - r(D_T(s) - \ell_{\text{cite}})$, with ΔI the index cost change ($\log_2 \frac{k+1}{k}$ per citation over k axioms). Up to ΔI , memorising at the first occurrence is cheaper iff $\beta < \beta^(s) := (D_T(s) - \ell_{\text{cite}})/|s|$.*

Remark 1.16 (Lemmas, compressible data, theorem streams). *[computed; known]*

pa §§3.3–3.5 (i) Commutativity costs 1358.6 bits with its two lemmas derived in line and 504.7 with them adopted (prior 90.5): lemmas used often become axioms (compare ??(c)). (ii) A schema compresses its direct instances about fourfold; for a non-primitive form, adopting $f(P)$ beats deriving and memorising after 1–2 uses. (iii) On streams of fresh induction uses (probability u) and library theorems, for $u = 0$ the MAP is $\mathbf{Q}$ plus the library from the first datum, a finite subtheory of PA and so strictly weaker (known; Ryll-Nardzewski, 1952); for $u = 0.1, 0.5$ it is $\mathbf{Q} + T_{\text{Ind}}$ plus the library, with PA’s theorems.

Proposition 1.17 (Gibbs). *[proved]^{pa Prop 3.2} If $\sum_s 2^{-\ell(s)} \leq 1$ (e.g. $\ell(s) = \beta|s|$) and P has finite entropy, then $\mathbb{E}_P[-\log_2 P] \leq \mathbb{E}_P[\ell]$. So on data from P_{T^*} the generator’s code length for a new datum is in expectation at most the prior cost of memorising it.*

Proposition 1.18 (Well specified, the generator wins). *[proved, given ?? and known facts; proof sketch (last part)]*

pa Prop 3.3 Let the data be i.i.d. from PA's L1 generator in setting W (??), weights positive and parameters admissible ($\text{supp } P_{\text{PA}} = \text{Th}(\text{PA})$, ??), in a countable class with $\pi(\text{PA}) > 0$. Then a.s. every $Q \cup S$ (S a finite set of PA-theorems) has posterior 0 eventually, and $\pi_n(C^*) \rightarrow 1$. Theories $\text{PA} \cup S$ (generic weights) decay polynomially, like spare slots.

Proposition 1.19 (“Must prove” with refutation). *[proved]*

pa Prop 3.4 Under S_{prove} with bounded derivability and R_d : (a) the posterior is the prior restricted to the score-1 theories; (b) $Q + T_{\text{Ind}}$ beats $Q \cup \{s_1, \dots, s_m\}$ once $\beta \sum |s_i| > 72$ bits (two library theorems suffice), if d covers its derivations; (c) any unrefuted theory with smaller prior deriving all data within d beats both, e.g. the bare formula metavariable F_0 , or an inconsistent theory whose shortest refutation exceeds d .

Answer. A derivation-length likelihood finds $Q + T_{\text{Ind}}$ from theorem data (1) when the data come from its own derivation process (Propositions 1.17 and 1.18), which human theorems, selected for short statements and interest, do not; (2) when they contain direct uses of the schema; (3) when the axiom prior is much steeper than the proof code ($\beta \geq \beta^*$); (4) under the 0/1 constraint with refutation, at the price of the size principle. Otherwise L1^{sch} memorises short theorems, and the MAP is strictly weaker than PA. L1^{eq} (??) and the fragment $\{Q4, Q5\}$ (??) show the same pattern: a derivation likelihood favours the theory that makes the data cheap to state. Not computed: theorem data under plain L1 or the full sum. A time bound on computing the likelihood sums over fewer derivations and so pushes further towards memorisation (??).

1.5 Data from $\text{Th}(\mathbb{N})$, the $\text{I}\Sigma_n$ chain and narrow practice

Proposition 1.20 (No consistent r.e. theory survives). *[proved]*

pa Prop 4.1 If $P_T(s) > 0$ iff $T \vdash s$ (e.g. L1, weights positive, parameters admissible) and the data are i.i.d. from a law with full support on $\text{Th}(\mathbb{N})$, every consistent r.e. theory has posterior 0 eventually, a.s.

Proposition 1.21 (Refutation). *[proved; known]*

pa Prop 4.2 Without a refutation rule inconsistent theories are never refuted; R_∞ refutes them at once but is not computable; R_{d_n} with $d_n \rightarrow \infty$ refutes each eventually; no computable rule does better uniformly, since consistency of r.e. theories is Π_1 -complete.

Along PA, $\text{PA} + \text{Con}(\text{PA})$, ... the posterior drifts; a toy model with logarithmic drift is in Remark A.2. In DT° each step must add a ground sentence:

Proposition 1.22 (Reflection is not a template). *[proved]* *pa Prop 4.3* Every DT° template whose instances lie in $\text{Ref}_T := \{\text{Prv}_T(\ulcorner \varphi \urcorner) \rightarrow \varphi\}$ (unary numerals) is ground. The proof is that of ?? with Prv_T for Accept_f .

Proposition 1.23 (Regret against a theory plus exceptions). *[proved]* *pa Prop 4.4* Under L_ε , for every T and D_n , with E the data T neither derives nor refutes: $-\log_2 P_{\text{mix}}(D_n) \leq -\log_2 \pi(T) + \sum_{s \notin E} -\log_2((1 - \varepsilon)P_T(s)) + \sum_{s \in E} -\log_2(\varepsilon\mu_0(s))$. With 0/1 support the same holds for $T \cup E$, the exceptions memorised at $\beta|e|$ prior bits each.

Pointwise limits are trivial: under 0/1 support memorisation alone eventually decides every sentence correctly (Remark A.1). So for $\text{Th}(\mathbb{N})$ what can be promised is regret against every theory plus its exceptions and depth-relative soundness, not identification. In Gold's model the chain $\text{I}\Sigma_1 \subset \text{I}\Sigma_2 \subset \dots$ with union PA is a limit point.

Proposition 1.24 (The $\text{I}\Sigma_n$ chain). *[(a) proved, given known facts; (b) proof sketch; (c) refuted]*

pa Prop 4.6; referee M4, m7

(a) For i.i.d. theorems from PA's L1 generator (setting W , parameters admissible), each $\text{I}\Sigma_n = \mathbf{Q} + \sigma_n$ has posterior 0 eventually and $\pi_n(C_{\text{PA}}^*) \rightarrow 1$ a.s. in any countable class with $\pi(\text{PA}) > 0$; the waiting times are presumably astronomical (a heuristic). (b) Before that, σ_n pays at each induction use for a Tarski biconditional that grows with the motive, so the schema wins linearly on induction-use data. (c) Refuted: “if the community uses only Σ_n motives, every DT° theory covering its practice contains PA-strength induction, so the posterior lands on the L_∞ side”.

(a) uses $\text{I}\Sigma_{n+1} \vdash \text{Con}(\text{I}\Sigma_n)$ (known; Hájek and Pudlák, 1993, location not verified), Gödel's second theorem and ???. (c) fails because AS Thm 5.16 concerns a single covering template and the wrappers of Proposition 1.14 need a formula metavariable (AS Thm 5.16 does hold with parameters, contrary to a remark of the referee). What is true, using that equivalent motives give equivalent induction instances (Lemma A.3):

Proposition 1.25 (Read-once induction templates). *[proved]^{pa Prop 4.8} Let $\tau = T_{\text{Ind}}[P := \lambda x.C]$ with C read-once: built by connectives and quantifiers from atoms $A(x, \bar{z}) = B(x, \bar{z})$ (A, B term metavariables) and $F(x, \bar{z})$ (F a formula metavariable), each metavariable occurring once. (a) If C has a formula metavariable, $\text{Q1} + \text{inst}(\tau) \vdash \text{Ind}(\psi)$ for every ψ . (b) Otherwise every instance has C 's formula skeleton, so $\text{Q} + \text{inst}(\tau) \subseteq \text{I}\Sigma_k$ for some k , and $\text{I}\Sigma_k \subsetneq \text{PA}$.*

The read-once condition is sufficient, not necessary (Remark A.4); which other templates give full induction is open.

Example 1.26 (Narrow practice ends on the weaker side). *[computed]^{pa §4.3 (1); referee r8} Practice: Q plus induction on motives $\exists z(s = t)$ and $\neg(s = t)$. $H_{\text{narrow}} := \text{Q} + \text{Ind}(\lambda x. \exists z A(x, z) = B(x, z)) + \text{Ind}(\lambda x. \neg C(x) = D(x))$ (term metavariables) covers it and lies in $\text{I}\Sigma_1 \subsetneq \text{PA}$. $\text{CL}(H_{\text{narrow}}) - \text{CL}(T^*) = +230.8, +212.5, +192.5, +174.6$ bits at $n = 100, 300, 1000, 3000$, matching Proposition 1.12(a) to 0.1 bit; slope $\frac{9-8}{2} - 3 \cdot \frac{9-1}{2} = -11.5$ bits per doubling (three fixed formula contexts), crossover $n \approx 2^{26.7}$.*

Example 1.27 (G1 keeps PA). *[computed; proof sketch (slope)]*

pa §4.3 (2)

u7's law G1 has 157 motive skeletons, each equivalent to a Σ_2 formula, so a term-only skeleton theory covers G1 inside $\text{I}\Sigma_2$; coverage does not force PA. But it is 42838.5 bits behind at $n = 3000$ (prior part 43040), with asymptotic slope $\frac{164-8}{2} - 19 \cdot \frac{9-1}{2} = +2.0$ bits per doubling (19 contexts; Wilks's theorem, from memory).

So deductive strength beyond usage is set by an Occam balance: a term-only theory gains $\frac{|A_c|-1}{2}$ bits per doubling for each context it fixes and pays $\frac{1}{2}$ per extra template. Narrow practice lands on the weaker side, broad practice on PA's; this is the Bayesian L_∞ -versus- L_5 choice (??).

Conjecture 1.28 (Broad practice). *[conjecture]^{pa Conj 4.10} If the motives have unbounded depth and use every connective at every depth with positive probability, the mass on theories deriving all of PA tends to 1. (Open because covering theories may use non-read-once templates.)*

Track experiments (fixed PCFG, L0, causal pools; language with $<$ and nine motive roots) finds the narrow effect with atomic motives. Let T_f be T_{Ind} with motive root f ; for a connective root, $\text{Q} + T_f$ has the same consequences as $\text{Q} + T_{\text{Ind}}$ (Proposition A.5), and Q3 is redundant given T_{Ind} (Proposition A.6).

candidate	$n = 10$	30	100	3000
root split (7 T_f)	+776.4	+773.6	+772.7	+768.3
spare + ($F \rightarrow F$)	+15.9	+16.6	+17.4	+19.8
memorise instances	+1792.3	+5392.7	+15901.1	+399693.6
over-general $Q + T_{L\infty}$ (AS Prop 5.12)	+549.5	+1500.6	+3832.5	+82717.0
lump $F_0 + T_{\text{Ind}}$	-174.7	-42.0	+319.3	+8929.8
bare F_0	+584.0	+2148.6	+6067.9	+137387.5

Table 3: Bayesian DTRC: $\text{CL}(\text{candidate}) - \text{CL}(Q + T_{\text{Ind}})$ in bits on G1 practice; one seed; hand-picked candidates. Computed (pa §5.2; all n in Appendix A.7).

Proposition 1.29 (Atomic fragments are weaker). *[proved, given two known facts]^{experiments Prop X10} $T^* := Q1..Q7 + T_{\text{Ind}}$ contains PA and proves $\sigma := \neg\exists x\exists y(\neg y = 0 \wedge x \cdot x = (SS0 \cdot y) \cdot y)$; $Q + \{T_-, T_<\} \not\vdash \sigma$ (known facts: $\text{PA} \vdash \sigma$; Shepherdson’s model, Shepherdson, 1964, secondary sources only).*

Remark 1.30 (E3(b)). *[computed]^{experiments E3(b)} L0 data from T^* with misspecified motive laws (5 seeds, $n \leq 2048$). Three connective-rich laws keep the theorems (mass 0.998–1.000 on T^* -equivalents at $n = 2048$); under two of them the MAP leaves T^* for $T^* + T_{\exists}$ or an observed-root split. Atomic motives: T^* has mass 0.988 at $n = 64$ in 4 of 5 seeds, then $Q + \{T_-, T_<\}$ has mass 1.000 in 5 of 5 seeds from $n = 1024$ (T^* -equivalents: $2 \cdot 10^{-178}$ at $n = 2048$): the posterior moves to a strictly weaker theory (??).*

1.6 Unlabelled mixtures: a Bayesian DTRC

Hypotheses: finite sets of DT° templates, prior 5 bits per symbol, Dirichlet($\frac{1}{2}$) weights, one shared positional grammar, negative data under R_0 . Data: G1 practice (Q1–Q7 as ground sentences, induction with probability $\frac{1}{2}$), one seed; hand-picked candidates (Table 3).

Proposition 1.31 (Spare slot). *[proved; computed]*

pa Prop 5.1 A never-used template τ added to K templates costs $\beta|\tau| + \frac{1}{2}\log_2 n + \log_2 \frac{\Gamma(K/2)}{\Gamma((K+1)/2)} + o(1)$ bits (??(a)); 19.03 and 19.82 bits at $n = 1000, 3000$ for $K = 8$, $\tau = F \rightarrow F$, against the computed 19.0, 19.8.

Without negatives $Q + T_{\text{Ind}}$ has mass $2.6 \cdot 10^{-53}$ at $n = 10$, $2.2 \cdot 10^{-13}$ at 30, and $\geq 1 - 6 \cdot 10^{-6}$ from $n = 100$; with two negatives ($0 = S0$ and a false instance of $T_{L\infty}$) $\geq 1 - 1.6 \cdot 10^{-5}$ from $n = 10$. The root split and the spare are PA-equivalent, so the mass on PA-equivalents is $\geq 1 - 10^{-96}$ from $n = 100$, *within this candidate set*. The split gains 1 bit per doubling (Proposition 1.12), the spare loses $\frac{1}{2}$; the rest lose linearly.

Remark 1.32 (Unsound lumps). *[computed]*

pa §5.4; experiments E2 At $n = 10, 30$ the unsound lump $F_0 + T_{\text{Ind}}$ leads by 174.7 and 42.0 bits (F_0 codes the rarely seen Q axioms through the grammar for less than seven ground templates cost), and $V_{\delta,0}$ accepts the false $0 = S0$ for every $\delta \geq 3 \cdot 10^{-13}$; the negative $0 = S0$ refutes the lump. This is AS’s “errs towards unsoundness for rarely used ground axioms” (AS App. E.7) in Bayesian form, and no contradiction with ???? ($\pi(Q + T_{\text{Ind}}) = 2^{-330}$ here; u7’s data are not from a model generator). One run in a hand-picked set; for the frequency see E2.

Remark 1.33 (E2). *[computed]*

experiments E2 L0 citations from T^ at fixed weights, 25 seeds, $n \leq 512$, causal pools. T^* takes the*

	what fails	evidence	fixable?
F1	usage over logic	Proposition 1.5 and Remark 1.7	change the target
F2	misspecified grammar: linear fragmentation	Table 6	shared positional grammar
F3	never-used parts split off (ω -gap)	Proposition 1.12 and Remark 1.34	harmless for theorems (Proposition 1.14)
F4	false generalisation, no counterexamples	Example 1.37	only by refutation
F5	unsound lumps at small n	Remarks 1.32 and 1.33	a negative datum, more data
F6	inconsistency unseen by the size principle	Proposition 1.36	only R_{d_n} , not certifiable
F7	no well-specified theory for $\text{Th}(\mathbb{N})$	Proposition 1.20 and Remark A.2	as regret (Proposition 1.23)
F8	strength beyond usage set by Occam terms	Example 1.26 and Remark 1.30	broad practice (Conjecture 1.28)
F9	bounded search memorises long proofs	Section 1.4	not in general (??)
F10	theorem data memorised, $\text{MAP} \subsetneq \text{PA}$	Proposition 1.15 and Remark 1.16	schema uses, steep prior

Table 4: Robust failures (pa §6); statuses as in the cited items.

mass once Q1, Q2, Q4–Q7 have each been cited (median datum 55); before that the sound but weaker $\text{SeenQ}(D_n)$ leads, a competitor pa did not score (data laws and candidate sets differ). Mean mass of T^* : 0.751 at $n = 64$, 0.996 at 512. A $\delta = 0.05$ verifier accepts a false probe in 5 of 25 seeds, at $n \leq 16$ (??). The complete root split stays about 700 bits behind; the false spare $T^* + \{0 = S0\}$ keeps 0.004 at $n = 512$ (??).

Remark 1.34 ($\forall x\varphi$ as a sound merge). [computed; identity proved]^{pa §5.5; referee M5} On data $t + 0 = t$ (t random closed terms) the schema $z + 0 = z$ wins (AS Prop 6.13). The sound split by the head of t covers every closed instance but does not entail $\forall x(x + 0 = x)$; it closes in at 5 bits per doubling (Proposition 1.12(c), $\frac{3-13}{2}$), crossover near $n = 10^8$: a Bayesian ω -gap (??), slower than in ?? because the parameter probability is learned. $\forall x(x + 0 = x)$ itself loses at a rate set by the rule code (Table 8): $\log_2 10$ bits per datum under the fixed code (the first version’s “ $2^{-3.3n}$, as H4 predicts” holds only for it), $\frac{R-1}{2} \log_2 n + O(1)$ under a learned depth-indexed code, $h(f)$ bits per datum in a practice with a fraction f of φ -instances (h the binary entropy). The direction holds under every code (??; universal’s learned shared c : ??).

Proposition 1.35 (“Does not contradict” learns nothing). [proved]

^{pa Prop 5.2} *If the likelihood is $\mu_0(s)[T \not\models \neg s]$ with μ_0 independent of T , the posterior is the prior restricted to compatible theories.*

L_ε is the minimal model that credits derivations without killing gaps, as Hänni asks (??(b)).

Proposition 1.36 (Inconsistency is invisible to the size principle). [proved]^{pa Prop 5.3} *Under $L1^{\max}$ with citation weights, adding σ with fixed weight fraction w gives $P_{T \cup \{\sigma\}}(s) \geq (1 - w)^{c(s)} P_T(s)$, $c(s)$ the citations in s ’s best T -derivation; with Dirichlet weights an unused σ costs its prior plus about $\frac{1}{2} \log_2 n$ bits. Consistency must be enforced by R_d .*

1.7 Robust failures and the overall answer

Example 1.37 (Euler’s polynomial (F4)). [computed; proof sketch (covering, eventual behaviour)]

^{pa §6 F4; referee m9} Data: $\text{Prime}(k \cdot k + k + 41)$, k geometric(0.9) conditioned on primality (false

k : 40, 41, 44, ...). The false schema $\text{Prime}(z \cdot z + z + 41)$ beats memorisation by 51445 bits at $n = 100$ and 314225 at $n = 10^6$ (flat grammar); no finite unguarded covering union is true without memorising. Under a depth-indexed grammar the predictive probability of $k = 40$ falls like $1/n$, yet the schema entails a falsehood; under the flat grammar memorisation should win beyond $n = 10^6$.

1.7.1 The overall answer

The inducer does *not* robustly pick up the textbook axioms; it picks up the axioms that are *used*. Within the hand-picked candidate sets, and for data that are direct uses of axioms, it puts nearly all mass on theories deductively equivalent to PA; in three of the four cases this is automatic, since every candidate is PA-equivalent (usage mixes, recursion conventions, root splits), and in the DTRC run PA-equivalents get mass $\geq 1 - 10^{-96}$ from $n = 100$. It fails robustly on theorem data, narrow practice, absent counterexamples, small n and inconsistency (Table 4). Not scored: all finite template unions, non-read-once templates, stronger true theories such as $\text{PA} + \text{Con}(\text{PA})$ (spare slots), further ZF candidates. The first version’s “it robustly gets a theory deductively equivalent to PA” is refuted as stated (referee M2–M4).

A Proofs and computations for Section 1

Proofs follow track pa’s final notes (and track experiments’ for Props X9–X11); steps they leave out are named. Track pa’s referee re-checked every derivation of Propositions 1.3, 1.8 and 1.14 and of Remark 1.9 with an independent checker, recomputed the cost tables, the toy tower and the Euler example by independent code, and read the proofs of Propositions 1.11, 1.20, 1.21, 1.23, 1.31, 1.35 and 1.36, Remark A.1 and Proposition 1.24(a), finding them correct for the stated models after the corrections recorded in ???. Added in the revision and therefore not refereed: the proof of Proposition 1.6, Proposition 1.12, Section 1.4 (Propositions 1.15 and 1.17 to 1.19), the full proof of Proposition 1.22, Lemma A.3, Proposition 1.25, Remarks A.4 and A.7, and experiments’ Propositions 1.29 and A.6. They were checked by the tracks and again while writing this appendix. Nothing is checked in a proof assistant.

A.1 The checker and the reproduction

`checks/nd.py` checks natural-deduction derivations line by line. Rules: hyp (open a hypothesis, to be discharged), ax (an instance of an axiom of the theory), tc (tautological consequence of earlier lines, by truth tables over their prime subformulas), $\rightarrow$ I, $\forall$ E (with a capture check), $\forall$ I and $\exists$ E (with eigenvariable conditions), $\exists$ I, refl and subst (equality). A derivation must end with no open hypotheses. Schematic derivations are written in a unary predicate symbol P ; substituting a formula for P , after renaming the derivation’s bound and eigenvariables away from it, maps derivations to derivations (the substitution theorem for predicate symbols; Shoenfield, 1967, location not verified). `test_nd.py` replays the four derivations of Proposition 1.3 on 200 random concrete motives with a parameter (seed 20261008): 200/200 check. Negative tests: the checker rejects all of 10 unsound steps (a $\forall$ I eigenvariable free in a hypothesis, a tc step that is not a tautology, an $\exists$ E eigenvariable that is not fresh in the conclusion or in a hypothesis, a mismatched subst, capture in $\forall$ E, a wrong goal, an open hypothesis, tc through a quantifier, a wrong $\exists$ I witness). The referee’s independent re-checker (own substitution, α -normal form, truth tables, rule semantics and statements of all axioms and schemas) accepts all 20 derivations of the

first version and rejects all 173 corrupted variants (a negated conclusion on some line, or a bad eigenvariable).

A script matched every number quoted in track pa’s notes against the outputs (87 quotes, 0 failures; three planted errors were caught). All twelve outputs were byte-identical on a second run. Seeds: c1 7, c2 `random.Random(n)` per n (as in u7), c3 1, c4 11 and 12, c5 5 and 6, c6 41 and 42, c8 31, 33, 11–13.

A.2 Equivalent axiomatisations

Proof of Proposition 1.3. Write “ φ is progressive” for $\forall x(\theta_\varphi \rightarrow \varphi)$. (A) Assume $\varphi(0)$ and $\forall x(\varphi \rightarrow \varphi(Sx))$; we show progressiveness. Let $\theta_\varphi(x)$. By Q3, $x = 0$, and then $\varphi(x)$; or $x = Sw$. In the second case $w + S0 = S(w + 0) = Sw$ by Q5 and Q4, so $w < x$ by Dlt; hence $\varphi(w)$, hence $\varphi(Sw)$, i.e. $\varphi(x)$. CVI(φ) gives $\forall x\varphi$. (B) Assume φ progressive and apply Ind to θ_φ . Base: $y < 0$ gives $y + Sz = 0$ for some z (Dlt), so $S(y + z) = 0$ by Q5, contradicting Q1; so $\theta_\varphi(0)$. Step: assume $\theta_\varphi(x)$ and $y < Sx$. Then $y + Sz = Sx$ for some z , so $S(y + z) = Sx$ (Q5) and $y + z = x$ (Q2). By Q3, $z = 0$ or $z = Sv$. If $z = 0$, then $y = x$ (Q4) and $\varphi(x)$ follows from $\theta_\varphi(x)$ by progressiveness. If $z = Sv$, then $y + Sv = x$, so $y < x$ and $\varphi(y)$ by $\theta_\varphi(x)$. So $\theta_\varphi(Sx)$, and $\forall x\theta_\varphi(x)$; progressiveness gives $\forall x\varphi$. (C1) If LNP(φ) fails, $\exists x\varphi$ holds and every x with $\varphi(x)$ has some $y < x$ with $\varphi(y)$. Then $\neg\varphi$ is progressive, so CVI($\neg\varphi$) gives $\forall x\neg\varphi$, a contradiction. (C2) Let φ be progressive and suppose $\exists x\neg\varphi$. LNP($\neg\varphi$) gives an x with $\neg\varphi(x)$ and $\forall y < x \varphi(y)$; progressiveness gives $\varphi(x)$, a contradiction. The base axioms cited by the checked derivations are exactly those named: (A) CVI, Dlt, Q3–Q5; (B) Dlt, Ind, Q1–Q5; (C1), (C2) only the schema. $\square$

For restricted motive classes the classical equivalences ($\text{I}\Sigma_n \Leftrightarrow \text{III}_n \Leftrightarrow \text{L}\Sigma_n \Leftrightarrow \text{LII}_n$; Kaye, 1991; Hájek and Pudlák, 1993, theorem numbers not checked) need bounded collection, because θ_φ adds a bounded quantifier, and some fail for parameter-free schemes (Kaye et al., 1988); DT° templates cannot say “motive in Σ_n ”, so only the full-schema equivalence matters here.

Proof of Proposition 1.5. $\text{CL}(\text{PA}_S; D_n) = \text{prior}_S + \sum_{i \leq n} (c_S(f_i) + L_Q(\varphi_i))$. The prior is a constant, and the L_Q terms are the same under every S . The differences $c_S(f_i) - c_{S'}(f_i)$ are i.i.d. and bounded, so the strong law gives the limit. The posterior odds are $2^{-\Delta^{\text{CL}}}$, which tend to 0 or ∞ exponentially when the limit is nonzero. $\square$

Proof of Proposition 1.6. For CVI(P); Ind(P) is the same with its 11-symbol antecedent (Ind(P) fails in the structure $\{0, a\}$ with $S0 = 0$, $Sa = a$, $P = \{0\}$, and it is not an instance of CVI, whose antecedent has root $\forall$). Both theories have 9 axioms, so a citation costs $\log_2 10 + \log_2 9 + 2\beta$ (it writes $P(x)$). The term positions of both forms lie under binders or are 0, so no $\forall E$ step can have produced them from a shorter variable without capture; formulas on the chain below have full size. Let π derive CVI(P) from PA_{Ind} . 1. π *cites an axiom*. CVI(P) is not valid: read $<$ as the full relation and P as empty; then $\forall y(y < x \rightarrow P(y))$ fails for every x (take $y = x$), so the antecedent holds, while $\forall xP(x)$ fails. So some line is a citation, costing at least $\log_2 10 + \log_2 9$. 2. *Another line writes at least 13 symbols*. Follow the premises back from the last line through $\forall E$ (premise $\forall v A$ with $A[t/v]$ the current formula), $\forall I$ (premise the matrix) and $\exists E$ (premise with the same conclusion). Every formula on this chain is CVI(P) with a quantifier prefix and terms in place of variables, which keeps the connective skeleton. The chain ends at a line of another rule. It is not ax: the Q axioms and Dlt do not contain P ; an Ind instance is an implication whose antecedent is a conjunction, while every chain formula, prefix removed, has an antecedent with root $\forall$; and no instance has a $\forall$ -prefix. It is not refl or $\exists I$ (wrong root). A hyp, tc or

subst line writes a formula with $\text{CVI}(P)$'s skeleton, at least 18 symbols (13 skeleton symbols and one at each of 5 term positions); a hyp line still writes it, even if discharged later. A $\rightarrow$ I line writes its discharged antecedent, which has the skeleton of $\text{CVI}(P)$'s antecedent: at least 13 symbols. So some non-citation line costs at least $\log_2 10 + 13\beta$. 3. Premise references cost ≥ 0 . So $\text{bits}(\pi) \geq 2\log_2 10 + \log_2 9 + 13\beta$, which exceeds the citation by $\log_2 10 + 11\beta$. $\square$

Proposition 1.8. Each law is proved by one induction: in T_{left} , $x + 0 = x$ on x ($0 + 0 = 0$; $Sx + 0 = S(x + 0) = Sx$) and $\forall y(x + Sy = S(x + y))$ on x ; in T_{right} , $0 + x = x$ on x and $Sp + x = S(p + x)$ on x with parameter p . The four derivations (11, 22, 11 and 21 lines) are checked; they cite only T_{Ind} and the other convention's two laws.

derivation (checked)	lines	symbols	writes of P	overhead	template prior
Sep from $\text{Repl}_J(\lambda xy. \varphi(x) \wedge y = x)$	36	207	19	1209.3	72 (Sep)
Repl_{sp} from Coll	15	108	12	574.1	185 (Repl_{sp})
Repl_J from Coll and two Sep instances	43	237	21	1432.0	167 (Repl_J)
Found from $\text{EInd}(\neg x \in S)$	21	138	0	758.1	109 (Found)

Table 5: ZF: overheads in bits per use (L1^{sch} , $\beta = \log_2 23$). Sep is $\forall X \exists Y \forall u(u \in Y \leftrightarrow u \in X \wedge \varphi)$; Repl_J is the image form $\forall x \forall y \forall z(\psi \wedge \psi[z/y] \rightarrow y = z) \rightarrow \forall X \exists Y \forall y(y \in Y \leftrightarrow \exists x(x \in X \wedge \psi))$; Repl_{sp} the bounding form with $\exists!$ spelled out; Coll, EInd, Found as in AS Table 6. The checker caught a capture error in a first version of the Sep constructor. Computed; upper bounds (pa §1.4).

Further numbers for Remark 1.7. On a 0.1 grid of $(p_{\text{Ind}}, p_{\text{CVI}})$, PA_{Ind} is the best minimal theory for $p_{\text{Ind}} \geq 0.7$, PA_{CVI} in a broad middle region and PA_{LNP} for large p_{LNP} ; PA_{all} , if allowed, wins every mixed cell. It pays $81 + 86$ prior bits and $\log_2(11/9) \approx 0.29$ bits per citation more than PA_{Ind} , and is ahead in expectation after 12, 2, 1, 8 schema uses for $(p_{\text{CVI}}, p_{\text{LNP}}) = (0.01, 0)$, $(0.05, 0)$, $(0.10, 0.05)$, $(0, 0.01)$; one CVI use alone saves 1468 bits. Under L1^{naive} the overhead grows by about $\beta(K - 1)$ bits per motive symbol, K the number of written copies of P : CVI from PA_{Ind} ($K = 14$) costs 1621, 1915, 2515, 3832 bits at $|\varphi| = 5, 10, 20, 40$ (schematic: 1468), and Ind from PA_{CVI} ($K = 18$) 1088, 1487, 2264, 4030 (schematic: 880); mean of 5 random motives per size, seed 7.

A.3 The MDL finding

Proof of Proposition 1.11. For an induction datum $\text{Ind}(\varphi)$ whose motive has root f , $P_{H_F} = w\mathcal{Q}(f)\mathcal{Q}(\varphi \mid f) = w\mathcal{Q}(\varphi) = P_{T^*}$; ground data have the same weight under both. $\square$

Proof of Proposition 1.12. (a) The total code is the prior, plus the KT code of the citation index, plus a sum over contexts of the KT code of the symbols coded there. Sequential KT in a context equals the Dirichlet($\frac{1}{2}$) marginal of that context's counts, whatever their order (known; it is the Pólya-urn form of the Dirichlet-multinomial; Krichevsky and Trofimov, 1981). By hypothesis the two hypotheses have the same count vectors in every context outside C , so those terms cancel; H' has no counts in C . The index and prior terms remain. (b) $\text{KT}(k; A) \ln 2 = \ln \Gamma(N + \frac{A}{2}) - \ln \Gamma(\frac{A}{2}) - \sum_{a: k_a > 0} [\ln \Gamma(k_a + \frac{1}{2}) - \ln \Gamma(\frac{1}{2})]$. Stirling, $\ln \Gamma(m + a) = (m + a - \frac{1}{2}) \ln m - m + \frac{1}{2} \ln 2\pi + o(1)$, applied to the $j+1$ growing arguments (j the number of nonzero counts) gives $N \ln N - \sum_a k_a \ln k_a + \frac{A-1}{2} \ln N + \frac{1-j}{2} \ln 2\pi + j \ln \Gamma(\frac{1}{2}) - \ln \Gamma(\frac{A}{2}) + o(1)$;

the first two terms are $N\hat{H}(k)$ in nats. (c) Let τ_0 be the template of H whose data visit c . H' 's counts are those of H for the other templates and $m_{c,a}$ for the templates that correspond to (τ_0, a) . Hence $n\hat{H}(n') = n\hat{H}(n) + m_c\hat{H}(m_c)$ (chain rule for empirical entropies), and the entropy terms of (a) cancel exactly. The $\log_2$ terms give $\frac{|H'| - 1}{2} - \frac{|H| - 1}{2} - \frac{|A_c| - 1}{2}$ times $\log_2 n$, since m_c is linear in n ; the rest is $O(1)$. For a split of T_{Ind} into the roots F , $|H'| - |H| = |F| - 1$ and $|A_c| = 9$. $\square$

Corrections and reproduction. The first version, like u7 itself, charged prior bits only for templates that some datum uses, while the KT index ranged over all seven (referee m1). Under G1 nothing changes; under G2 (no $\vee, \forall, \exists$ at the root) 355 prior bits were missing and under G3 470. Every template is now charged; H_{used} is AS's split H_F with the index restricted to F . With u7's own bookkeeping the script reproduces AS Table 12 exactly at $n = 1000$ (G1: $-223, +2325, +2701$; G2: $-915, +1141, +1428$); the G2 rows of AS Table 12 carry the same undercount, which changes no conclusion there. A first SDPC run ($+803$ to $+12270$ bits) was an encoding artefact and is refuted.

law, split	n	NAIVE	PC	DPC	SDPC	RDPC	CF
G1, H_{root}	1000	-223.0	+2325.3	+2700.6	+770.0	+1676.6	+709.8
	16000	-16466.1	-1198.5	-3035.6	+765.8	+3724.6	-301.8
	256000	-274655.9	-94088.5	-163027.5	+761.9	+5985.9	-17241.7
G2, H_{used}	1000	-926.0	+1129.8	+1417.1	+403.7	+870.1	+236.5
	16000	-21484.6	+1035.8	+2805.3	+393.7	+1566.1	-2341.2
	256000	-350773.1	-12391.6	+4289.5	+383.7	+2346.1	-45211.8
G3, H_{used}	1000	-1300.8	+772.5	+543.0	+284.6	+931.3	+270.0
	16000	-24985.6	-3058.9	-9258.6	+272.8	+1736.6	-246.7
	256000	-404445.7	-75061.9	-186306.5	+260.8	+2564.9	-7814.9

Table 6: $\text{CL}(\text{split}) - \text{CL}(T^*)$ in bits ($T^* = \mathbf{Q} + T_{\text{Ind}}$; positive: T_{Ind} stays whole). Prior parts $+780$ (G1), $+425$ (G2), $+310$ (G3). SDPC for H_{root} under G2 and G3: $+769.9 \rightarrow +761.8$ and $+769.7 \rightarrow +761.9$. CF on G1 at $n = 256000$: -0.067 bits per datum. Computed (pa §2.2; c2_md1.out, c2b_cf.out).

Proof of Proposition 1.14. (a) Each wrapper $w_f(P)$ ($\neg\neg P, P \wedge P, P \vee P, (0 = 0) \rightarrow P, \forall z P, \exists z P$, with z not free in P) is logically equivalent to P and has root f , so $\text{Ind}(w_f(P)) \in \text{inst}(T_f)$, and $\text{Ind}(P)$ follows by Lemma A.3 (a special case of Proposition 1.25(a)). The six checked derivations have 17–19 lines; overheads over citing $\text{Ind}(P)$: $\neg$ 409.9, $\wedge$ 441.6, $\vee$ 441.6, $\rightarrow$ 488.9, $\forall$ 369.2, $\exists$ 425.6 bits. No wrapper exists for the root $=$. (b) The instances of $T_{=}$ are induction for equations, so $\mathbf{Q} + T_{=} \subseteq \text{IOpen}$; IOpen does not prove that $\sqrt{2}$ is irrational (Shepherdson, 1964), and PA does. $\square$

A.4 Theorem data

Proofs of Propositions 1.15 and 1.17. Proposition 1.15: adding s adds $\beta|s|$ prior bits; each occurrence of s costs ℓ_{cite} instead of $D_T(s)$; the other data change only through the index. Proposition 1.17: let $Z := \sum_s 2^{-\ell(s)} \leq 1$ and $q := 2^{-\ell}/Z$. Then $\mathbb{E}_P[\ell + \log_2 P] = \sum_s P(s) \log_2 \frac{P(s)}{Zq(s)} = \text{KL}(P||q) - \log_2 Z \geq 0$, with equality iff $P = q$ and $Z = 1$. Adopting a new datum costs at least $\ell(s)$ prior bits, so memorisation can gain only on repeated data. $\square$

Proof of Proposition 1.18. $\text{Th}(\mathbf{Q} \cup S) \subseteq \text{Th}(\mathbf{PA})$, and equality would make $\mathbf{PA}$ finitely axiomatisable, which it is not (Ryll-Nardzewski, 1952). So some $\sigma \in \text{Th}(\mathbf{PA}) \setminus \text{Th}(\mathbf{Q} \cup S)$ has $P_{\mathbf{PA}}(\sigma) > 0$ (??, parameters admissible) and $P_{\mathbf{Q} \cup S}(\sigma) = 0$; it occurs a.s. after finitely many data. There are countably many such S . The second claim is ???. For $\mathbf{PA} \cup S$: for generic weights its law differs from $P_{\mathbf{PA}}$, so it loses by ???; that the decay is polynomial, as for a spare slot (??), is a proof sketch, and whether some special weight gives $P_{\mathbf{PA} \cup S} = P_{\mathbf{PA}}$ was not checked. $\square$

Proof of Proposition 1.19. (a) is the definition of the score. (b) and (c) compare prior bits among score-1 theories: F_0 has every sentence as an instance, and an inconsistent theory derives every sentence, so either has score 1 unless refuted within d . The example in (b): $\beta(7 + 13) = 31.7 + 58.8 = 90.5 > 72$ bits. $\square$

The theorems in other axiomatisations. The six theorems of Table 2 cost 1148.4–4163.7 bits more in $\mathbf{PA}_{\text{CVI}}$ and 2241.2–8117.4 bits more in $\mathbf{PA}_{\text{LNP}}$ than in $\mathbf{Q} + T_{\text{Ind}}$, with $\beta^* = 143$ –501 and 240–861 bits per symbol (upper bounds: Ind is re-derived at each use; a direct $\mathbf{PA}_{\text{CVI}}$ derivation of $\forall x 0 + x = x$ by the referee costs 652 bits more instead of 1161). The four recursion-convention theorems of Proposition 1.8 give $\beta^* = 30.5$ –45.6. Decodable-code costs are 3.6–7.5% higher. All 19 derivations are checked.

Compressible data and streams (Remark 1.16). Per datum $f(\varphi)$ (φ a random motive, seed 41) the options cost: derive in $\mathbf{PA}_{\text{Ind}}$, memorise $(\beta|f(\varphi)| + \ell_{\text{cite}})$, or adopt $f(P)$ ($\ell_{\text{cite}} + \beta|\varphi|$, plus $\beta|f(P)|$ once). Ind, $|\varphi| = 21$: 101.5 (a citation) against 431.7. CVI, $|\varphi| = 20$: derive 1565.0, memorise 332.2, adopt 97.0 (+81); $|\varphi| = 418$: 3365.4, 5733.3, 1897.3. LNP, $|\varphi| = 21$: 2335.5, 350.3, 101.5 (+86). Streams (seed 42; bits relative to $\mathbf{Q} + T_{\text{Ind}}$, $n = 1000$): $u = 0$: $\mathbf{Q} + T_{\text{Ind}}$ plus library -561857 , $\mathbf{Q}$ plus library -562036 (MAP); $u = 0.1$: -513380 (MAP) against -489898 ; $u = 0.5$: -285950 (MAP) against -147800 .

A.5 Data from $\text{Th}(\mathbb{N})$, the chain and narrow practice

Proof of Proposition 1.20. If T proves a false φ , then $\neg\varphi \in \text{Th}(\mathbb{N})$ and $T \not\vdash \neg\varphi$ by consistency. Otherwise T 's theorems are true, and since they are r.e. while $\text{Th}(\mathbb{N})$ is not, some true σ has $T \not\vdash \sigma$. In both cases some s has $\mu(s) > 0 = P_T(s)$, and the probability that it is absent from n draws is $(1 - \mu(s))^n \rightarrow 0$. $\square$

Proof of Proposition 1.21. An inconsistent theory derives every sentence, so its 0/1 likelihood is never 0 without a rule. Under R_∞ it derives the negation of the first datum. Under R_{d_n} its shortest refutation has some finite size d , and it is refuted once $d_n \geq d$. Consistency of r.e. theories is Π_1 -complete (known), so no computable rule refutes exactly the inconsistent theories. (AS Thm 6.15 is a related but different statement about DTRC-type accept sets.) $\square$

Proof of Proposition 1.22 (adapted from the proof of ???). Suppose τ has a metavariable M with an occurrence at position q . Constant bodies of each sort with two different root symbols exist: $\lambda\bar{z}.0$ and $\lambda\bar{z}.(0 + 0)$ for terms, $\lambda\bar{z}.(0 = 0)$ and $\lambda\bar{z}.\neg(0 = 0)$ for formulas; for a constant body the subtree of $\tau\theta$ at q is that constant (AS Lemma 2.4), and the symbols of τ above q are kept. Give every other metavariable any body. *Case 1: q is the root or lies in the antecedent.* If q lies inside the numeral argument of Prv_T , take $\lambda\bar{z}.(0 + 0)$: the argument is then not a numeral, so $\tau\theta \notin \text{Refl}_T$. Otherwise q is the root or a position of Prv_T 's fixed skeleton, where all members of

Refl_T carry the same symbol; choose the constant body with a different root symbol. *Case 2: no metavariable occurs at the root or in the antecedent.* Then the antecedent is a fixed $\text{Prv}_T(N_0)$, and every member of Refl_T with this antecedent is $\text{Prv}_T(\ulcorner \varphi_0 \urcorner) \rightarrow \varphi_0$ for the one φ_0 with $\ulcorner \varphi_0 \urcorner = N_0$; so $\text{inst}(\tau)$ has at most one element. But a template with a metavariable has infinitely many instances, since each type has infinitely many bodies and $\theta \mapsto \tau\theta$ is injective (AS Thm 2.5(a)). So τ has no metavariable. $\square$

Remark A.1 (Pointwise limits are trivial). [proved]

pa Rem 4.5 Under 0/1 support every sentence is eventually decided correctly by memorisation alone (with inconsistent theories present, given R_{d_n}). Whether the mass of not-yet-refuted inconsistent theories deriving a sentence tends to 0 is open.

Proofs of Proposition 1.23 and Remark A.1. $P_{\text{mix}}(D_n) \geq \pi(T) \prod_s L_\varepsilon(T; s)$; for $s \notin E$ that T derives, $L_\varepsilon(T; s) \geq (1 - \varepsilon)P_T(s)$ (if T refutes s the bound is $+\infty$); for $s \in E$, $L_\varepsilon(T; s) \geq \varepsilon\mu_0(s)$. Take $-\log_2$. With $\varepsilon = 0$, apply the same to $T \cup E$, whose likelihood is positive on all of D_n . For Remark A.1: a true s occurs eventually, after which every surviving theory derives it; for a false s , $\neg s$ occurs eventually, after which no surviving consistent theory derives s ; with inconsistent theories present this needs R_{d_n} (Proposition 1.21). Their remaining mass can be large (Proposition 1.36); whether it tends to 0 is open. $\square$

Remark A.2 (Toy drift). [computed]

pa §4 Along $T_0 = \text{PA}$, $T_{k+1} = T_k + \text{Con}(T_k)$, with Geometric($\frac{1}{2}$) data levels and prior $2^{-50(k+1)}$, the MAP is the largest level seen (16 at $n = 10^5$) and the regret grows like 50 bits per level (798.9 bits at $n = 10^5$; linear, 0.2637 bits per datum, if the theories' level law is misspecified). The toy covers only the provable part of the data: under a full-support law a positive fraction of the data is unprovable in every T_k . Each step adds a ground sentence, since one reflection sentence suffices (??) and the reflection schema is not a template (Proposition 1.22).

Details (seed 1; data levels Geometric(ρ), each T_k generating its theorems with a truncated geometric level law of rate s): for $\rho = s = \frac{1}{2}$ the MAP level and regret are 2, 98.1 bits at $n = 10$, 9, 448.6 at 10^3 and 16, 798.9 at 10^5 , and the probability that the next datum is unprovable is $7.6 \cdot 10^{-6}$ at 10^5 ; with L_ε , $\varepsilon = 0.01$, the MAP lags the record (11 at 10^5) and the regret is 674.0; for $\rho = s = 0.8$ the regret is 2598.9 bits at 10^5 (MAP 52); for $\rho = \frac{1}{2}$, $s = 0.3$ it is 0.2637 bits per datum at 10^5 . The referee's independent re-implementation gives identical numbers.

Proof of Proposition 1.24. (a) $I\Sigma_n$ is finitely axiomatisable for $n \geq 1$ (known). $I\Sigma_{n+1} \vdash \text{Con}(I\Sigma_n)$ (known; Hájek and Pudlák, 1993, traced by the referee to Cor. I.4.34 through a secondary source; not verified), so $\text{PA} \vdash \text{Con}(I\Sigma_n)$, while $I\Sigma_n \not\vdash \text{Con}(I\Sigma_n)$ by Gödel's second theorem, $I\Sigma_n$ being consistent (true). So $P_{\text{PA}}(\text{Con}(I\Sigma_n)) > 0 = P_{I\Sigma_n}(\text{Con}(I\Sigma_n))$, and $\text{Con}(I\Sigma_n)$ occurs a.s. Eliminating each $I\Sigma_n$ is not concentration, since the tail $\{I\Sigma_m : m > N\}$ could hold mass; ?? gives $\pi_n(C_{\text{PA}}^*) \rightarrow 1$, and no $I\Sigma_n$ is in C_{PA}^* since its law has a different support. The waiting time is about $1/P_{\text{PA}}(\text{Con}(I\Sigma_n))$. (b) *Sketch.* Deriving $\text{Ind}(\varphi)$, $\varphi \in \Sigma_n$, from σ_n goes through the Tarski biconditional $\varphi \leftrightarrow \text{Tr}_n(\ulcorner \varphi \urcorner)$ (the predicate of ??), whose derivation grows with $|\varphi|$, while citing T_{Ind} costs $L_Q(\varphi)$ plus a constant. Not formalised; the growth rate was not checked. (c) Refuted by Examples 1.26 and 1.27. $\square$

Lemma A.3 (Equivalent motives). [proved]

pa Lemma 4.7 If $B \vdash \forall \vec{p} \forall x (\chi \leftrightarrow \psi)$ then $B + \text{Ind}(\chi) \vdash \text{Ind}(\psi)$.

Proof of Lemma A.3. From $\forall x(\chi \leftrightarrow \psi)$ at 0, x and Sx : $\psi(0)$ gives $\chi(0)$, and $\psi(x) \rightarrow \psi(Sx)$ gives $\chi(x) \rightarrow \chi(Sx)$. So the antecedent of $\text{Ind}(\psi)$ gives that of $\text{Ind}(\chi)$, which gives $\forall x\chi$, hence $\forall x\psi$. Generalise over the parameters. $\square$

Proof of Proposition 1.25. (a) Fix the path from C 's root to one occurrence $F(x, \bar{z})$. Instantiate every other atom as a constant: $\top := (0 = 0)$ ($A, B := \lambda.0$), $\perp := (S0 = 0)$ ($A := \lambda.S0$, $B := \lambda.0$), other formula metavariables $\lambda.(0 = 0)$ or $\lambda.(S0 = 0)$; Q1 refutes $S0 = 0$. Since each metavariable occurs once, these choices are independent, so each sibling on the path can be made $\top$ or $\perp$ as needed: $\top$ beside $\wedge$ and $\leftrightarrow$, $\perp$ beside $\vee$, $\top$ as the antecedent of $\rightarrow$ when the path goes to the consequent, and $\perp$ as the consequent when it goes to the antecedent (that step then acts as a negation, as $\neg$ does). Set $F := \lambda x \bar{z}. \psi'(x)$, which ignores $\bar{z}$, so the quantifiers on the path become vacuous. By propositional logic, Q1 and the vacuous-quantifier laws, $\text{Q1} \vdash \forall x(C\theta \leftrightarrow \psi')$ or $\text{Q1} \vdash \forall x(C\theta \leftrightarrow \neg\psi')$, depending only on the path. Take $\psi' := \psi$ in the first case and $\psi' := \neg\psi$ in the second. Then $\text{Q1} \vdash \forall x(C\theta \leftrightarrow \psi)$, and Lemma A.3 applied to the instance $\tau\theta = \text{Ind}(C\theta)$ gives $\text{Ind}(\psi)$. (b) Term metavariables are replaced by terms, which contain no logical symbols, so every instance has C 's skeleton. A formula with a fixed skeleton is logically equivalent to a prenex formula of fixed complexity Σ_k , and Lemma A.3 turns the instances into Σ_k induction. $\text{I}\Sigma_k \subsetneq \text{PA}$ because $\text{PA} \vdash \text{Con}(\text{I}\Sigma_k)$ (known, as in Proposition 1.24(a)) while $\text{I}\Sigma_k \not\vdash \text{Con}(\text{I}\Sigma_k)$. $\square$

Remark A.4 (Read-once is not necessary). [proved; computed]

pa Rem 4.9 $T_{\text{Ind}}[P := \lambda x. x = 0 \vee F(x)]$ is not read-once, and every instance's motive holds at 0; yet $\text{Q1}, \text{Q2} + \text{Ind}(x = 0 \vee \exists y(x = Sy \wedge P(y))) \vdash \text{Ind}(P)$ (checked, 47 lines). Which non-read-once templates give full induction is open.

The checked derivation has 227 written symbols, costs 1403.3 bits against a 15.4-bit citation, and cites only Ind , Q1 , Q2 . The motive χ is not equivalent to P , so Lemma A.3 does not apply.

Details of Examples 1.26 and 1.27. Narrow practice (the referee's counterexample r8, rechecked by the track): $|T_{\text{Ind}}| = 14$, $|T_E| = 39$, $|T_N| = 31$ (`dtlib`), prior difference 280 bits; coverage checked on 3000 data in seeds 31 and 33; the code, the identity (three contexts: motive root and the two “=” positions) and its expansion give +230.8/ +230.8/ +232.0 at $n = 100$ and +174.6/ +174.6/ +174.6 at 3000. G1: motives of depth 2 quantifying only atoms give $1 + 7 + 3 \cdot 49 + 2 = 157$ skeletons, all found among 200000 motives (seed 12); 112 occur in the first 3000 data. Each motive is a Boolean combination of atoms and single quantifiers over atoms, so it is equivalent to a Σ_2 formula. The theory of the 112 seen skeletons (119 templates) is +42905.6 and +42838.5 bits behind at $n = 1000, 3000$; T^* uses 19 formula contexts. With all 157 skeletons (164 templates) the slope is +2.0 bits per doubling. The entropy terms cancel only asymptotically: G1 draws each connective independently given its context, so their difference is $O_P(1)$ by Wilks's theorem for nested multinomial models (cited from memory; proof sketch).

A.6 Root fragments and the Q axioms

Track experiments' $T^* = \text{Q1}.. \text{Q7} + T_{\text{Ind}}$ has one-hole motives without parameters; its language has $<$ with no Q axiom about it. T_f for $f \in \{=, <\}$ has motive $a(x) f b(x)$ with 1-ary term metavariables; for $\neg$, $\neg P_1(x)$; for $\wedge, \vee, \rightarrow, \leftrightarrow$, $P_1(x) f P_2(x)$; for $\forall, \exists$, $f y P_1(x, y)$.

Proposition A.5 (Root fragments). [proved]^{experiments Prop X9} $\text{inst}(\{T_f : 9 \text{ roots}\}) = \text{inst}(T_{\text{Ind}})$; for each connective root f and any Q^- , $\text{Q}^- \cup \{T_f\}$ and $\text{Q}^- \cup \{T_{\text{Ind}}\}$ have the same consequences; the argument fails for $=$ and $<$.

Proof of Proposition A.5. (a) Every motive has one of the 9 roots, and an instance of T_{Ind} with root f is the instance of T_f whose sub-bodies are read off the motive (for $\forall, \exists$ the bound variable becomes P_1 's second argument); conversely $\text{inst}(T_f) \subseteq \text{inst}(T_{\text{Ind}})$. (b) One direction is (a). For the other, given φ choose: $P_1 = P_2 = \varphi$ for $\wedge, \vee$; $P_1 = \neg\varphi$ for $\neg$; $P_1 = (0 = 0)$, $P_2 = \varphi$ for $\rightarrow, \leftrightarrow$; $P_1(x, y) = \varphi(x)$ for $\forall, \exists$ (nonempty domains). The chosen motive ψ satisfies $\vdash \forall x(\psi \leftrightarrow \varphi)$, so $\text{Ind}(\psi) \vdash \text{Ind}(\varphi)$. (c) No such body exists for an atomic root; Proposition 1.29 shows the atomic fragments are weaker. $\square$

Proof of Proposition 1.29. T^* proves every induction instance with parameters. For $\varphi(x, \bar{w})$ take the parameter-free motive $\psi(x) := \forall \bar{w}[(\varphi(0, \bar{w}) \wedge \forall y(\varphi(y, \bar{w}) \rightarrow \varphi(Sy, \bar{w}))) \rightarrow \varphi(x, \bar{w})]$. $\psi(0)$ is valid; if $\psi(x)$ and the base and step hold for $\bar{w}$, then $\varphi(x, \bar{w})$ and so $\varphi(Sx, \bar{w})$: $\psi(Sx)$. T_{Ind} gives $\forall x\psi$, which is induction for φ . So $T^* \supseteq \text{PA}$ and $T^* \vdash \sigma$, as $\text{PA} \vdash \sigma$ (known). $\mathbf{Q} + \{T_=: T_<\} \not\vdash \sigma$. By Shepherdson (1964) (secondary sources only) there is a model M of $\mathbf{Q}$ plus open induction with parameters in which $x \cdot x = 2 \cdot y \cdot y$ has a solution with $y \neq 0$; M is the non-negative part of a discretely ordered ring, with $Sx := x + 1$, so Q1–Q7 hold (Q3 by discreteness). Instances of $T_=:$ are open induction, so they hold. Read $<$ as the empty relation: every instance of $T_<$ has the false conjunct $a(0) < b(0)$ in its antecedent, so it holds; σ does not mention $<$. So $M \models \mathbf{Q} + \{T_=: T_<\}$ and $M \not\models \sigma$. $\square$

The same holds for any theory whose induction-type templates are all term-only: by Proposition 1.25(b) it lies in some $\text{IS}_k \subsetneq \text{PA}$.

Proposition A.6 (Redundant Q axioms). *[proved; computed]*

experiments Prop X11 $\text{Q1, Q2, Q4..Q7} + T_{\text{Ind}} \vdash \text{Q3}$; no other Q_i is redundant given full induction.

Proof of Proposition A.6. T_{Ind} cites $\text{Ind}(\psi)$ for $\psi(x) := x = 0 \vee \exists y(x = Sy)$; $\psi(0)$ holds and $\psi(Sx)$ holds with $y := x$, so $\forall x\psi$, which gives Q3. For the other axioms: in a structure in which every element is the value of a numeral, every induction instance holds (induction on k in the metatheory). Each structure below has this property, satisfies the Q_j , $j \neq i$, and falsifies Q_i : $i = 1$: $\{0\}$, all operations 0; $i = 2$: $\{0, 1\}$, $S0 = S1 = 1$, $x + 0 = x$, $x + 1 = 1$, $x \cdot 0 = 0$, $x \cdot 1 = x$; $i = 4$: $\mathbb{N}$, $x \oplus y := x + y + 1$, $x \otimes y := y(x + 1)$; $i = 5$: $\mathbb{N}$, $x \oplus y := x$, $x \otimes y := 0$; $i = 6$: $\mathbb{N}$, standard $+$, $x \otimes y := xy + 1$; $i = 7$: $\mathbb{N}$, standard $+$, $x \otimes y := 0$. Each verification is a line of arithmetic (e.g. $i = 4$: $x \oplus Sy = x + y + 2 = S(x \oplus y)$, $x \otimes Sy = (y + 1)(x + 1) = (x \otimes y) \oplus x$); `check_models.out` evaluates all seven axioms in each structure on a finite range. $\square$

A.7 The Bayesian DTRC, the $\forall$ -merge and Euler's polynomial

Proof of Proposition 1.31. The Dirichlet($\frac{1}{2}$) marginal of counts $(n_1, \dots, n_K, 0)$ over $K + 1$ categories divided by that of $(n_1, \dots, n_K)$ over K is $\frac{\Gamma((K+1)/2)\Gamma(n+K/2)}{\Gamma(K/2)\Gamma(n+(K+1)/2)}$, and $\Gamma(n+a+\frac{1}{2})/\Gamma(n+a) \sim n^{1/2}$. Add $\beta|\tau|$ prior bits. $\square$

Proofs of Propositions 1.35 and 1.36. Proposition 1.35: the likelihood is the constant $\mu_0(D)$ for every compatible theory and 0 otherwise. Proposition 1.36: every T -derivation is a T' -derivation, and each of its citations loses a factor $1 - w$ under the rescaled weights; the best T' -derivation is at least as probable. With Dirichlet weights and σ unused, Proposition 1.31 applies. $\square$

Remark A.7 (Overlapping templates). *[proof sketch]*

pa Rem 5.4 For the lump and F_0 the code assigns each datum to its cheapest covering template, a

two-part bound; if the greedy assignment is optimal (not proved), the exact marginal is at most about n bits lower, and the lump stays ≥ 219 bits behind at $n = 100$.

Sketch for Remark A.7. The exact marginal sums over all assignments of data to covering templates. Its code length is at least the body code length under the best assignment, because the index probabilities of all assignments sum to at most 1. If the best assignment is the greedy one (induction data on T_{Ind} , whose motive is coded once, rather than on F_0 , which codes the whole sentence), the exact code is lower than the two-part code by at most the greedy index cost, about n bits for two covering templates. At $n = 100$ the lump would then still be at least $319.3 - 100 \approx 219$ bits behind. Optimality of the greedy assignment is not proved.

Full dtrc run. Table 3 at $n = 300, 1000$: root split +771.4, +769.9; spare +18.2, +19.0; memorise +41418.2, +133845.4; over-general +8964.3, +27876.8; lump +1048.9, +3205.5; bare F_0 +14712.6, +46251.0 bits (the first version had eight entries that did not match the output; referee M1). Without negatives the lump has mass 1.000 at $n = 10$ and $7.679 \cdot 10^{-97}$ at $n = 100$; from $n = 100$ the remainder beside $Q + T_{\text{Ind}}$ is almost all on the spare ($5.808 \cdot 10^{-6}$). The second negative is $(0 = 0) \wedge \forall x(0 = S0 \rightarrow 0 = S0) \rightarrow \forall x(0 = S0)$. The theories without $0 = S0$ as an instance have mass $2.6 \cdot 10^{-53}$ and $2.2 \cdot 10^{-13}$ at $n = 10, 30$, hence the acceptance by $V_{\delta,0}$ for $\delta \geq 3 \cdot 10^{-13}$.

candidate	$n = 10$	30	100	300	1000	3000
split by head of t (sound)	+115.5	+109.1	+101.0	+93.3	+84.6	+76.7
memorise instances	+519.5	+1263.8	+3749.9	+12204.1	+34023.6	+83197.3
over-general $z_1 + 0 = z_2$	+145.5	+351.5	+927.2	+2618.8	+7975.5	+22702.2
$\forall x(x + 0 = x)$, fixed rule code	+38.2	+104.7	+337.2	+1001.6	+3326.9	+9970.8

Table 7: $\forall x\varphi$ as a sound merge: $\text{CL} - \text{CL}(z + 0 = z)$ in bits on data $t + 0 = t$ (t random closed terms, seed 12). The over-general template is refuted by the negative $0 + 0 = S0$. With Q3 in the theory the head split would give $\forall x(x + 0 = x)$ by cases; here the theories are the templates alone. A first run in which the head split overtook the schema at $n = 1000$ was an artefact of coding split bodies from a root context (refuted). Computed (pa §5.5).

n	$R = 10$ (rules of ND)			$R = 2$ ($C_{\min}$: cite, $\forall E$)		
	fixed	learned, shared	learned, by depth	fixed	shared	by depth
10	33.2	23.4	12.5	10.0	20.0	2.5
10^3	3321.9	2004.0	41.1	1000.0	2000.0	5.8
10^6	3321928.1	2000004.0	85.9	1000000.0	2000000.0	10.8

Table 8: $\text{CL}(H_{\forall}) - \text{CL}(H_{\text{sch}})$ in bits on pure instance data under three rule codes (term costs cancel). The depth-indexed code gives $\frac{R-1}{2} \log_2 n + O(1)$ ($4.31 \log_2 n$ at $n = 10^6$ for $R = 10$, tending to 4.5). In a mixed practice with a fraction f of φ -instances the depth-indexed difference is $h(f)$ bits per datum at $n = 10^6$ (0.4691 at $f = 0.9$ and 0.1; 1.0001 at $f = \frac{1}{2}$), the shared one 1.896, 1.377, 0.483 bits per datum at $f = 0.9, 0.5, 0.1$. Computed in closed form; matches the referee's r3 (pa §5.5).

Euler’s polynomial (Example 1.37). $\text{Prime}(m) := \neg m = 0 \wedge \neg m = S0 \wedge \forall a \forall b (a \cdot b = m \rightarrow a = S0 \vee b = S0)$ has a 27-symbol frame and three occurrences of $m = k \cdot k + k + 41$ (the first version counted two; referee m9), so $|\text{datum}(k)| = 27 + 3(3(k + 1) + 45)$ with unary numerals; the template has 171 symbols. Under the unconditioned geometric law the false instances have mass 0.0049 ($\rho = 0.9$) and 0.0727 ($\rho = 0.97$). Code length of memorisation minus the false schema (bits), flat / depth-indexed grammar, and the schema’s predictive probability of $k = 40$: $\rho = 0.9$, $n = 100$: 51445/51373, $1.9 \cdot 10^{-3}/4.1 \cdot 10^{-4}$; $n = 10^4$: 161030/160816, $1.4 \cdot 10^{-3}/4.8 \cdot 10^{-5}$; $n = 10^6$: 314225/320382, $1.4 \cdot 10^{-3}/5.0 \cdot 10^{-7}$; $\rho = 0.97$, $n = 10^6$: 1820112/1920893, $8.6 \cdot 10^{-3}/5.0 \cdot 10^{-7}$ (seeds 5, 6; primality by trial division; the referee’s closed-form recomputation is identical). *Covering (sketch).* A template that covers infinitely many data must have a metavariable at or above a position of the numeral k (otherwise k is fixed), so some instance puts there a term that makes the sentence false: the numeral $k = 41j$ at the positions of k ($41 \mid k^2 + k + 41$), or a composite such as $41 \cdot 41$ at a position above them. So every finite unguarded covering union is false unless it memorises. *Eventual behaviour (sketch).* Under the flat grammar the schema pays a linear KL term (about 0.007 bits per datum at $\rho = 0.9$) for the false k it predicts, while memorisation’s prior grows like $(\log n)^2$, so memorisation eventually wins, beyond $n = 10^6$; under the depth-indexed grammar both costs grow like $(\log n)^2$ and the false schema stays ahead. Constants estimated, not computed. A refutation datum ($40^2 + 40 + 41 = 41^2$) removes the schema.

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